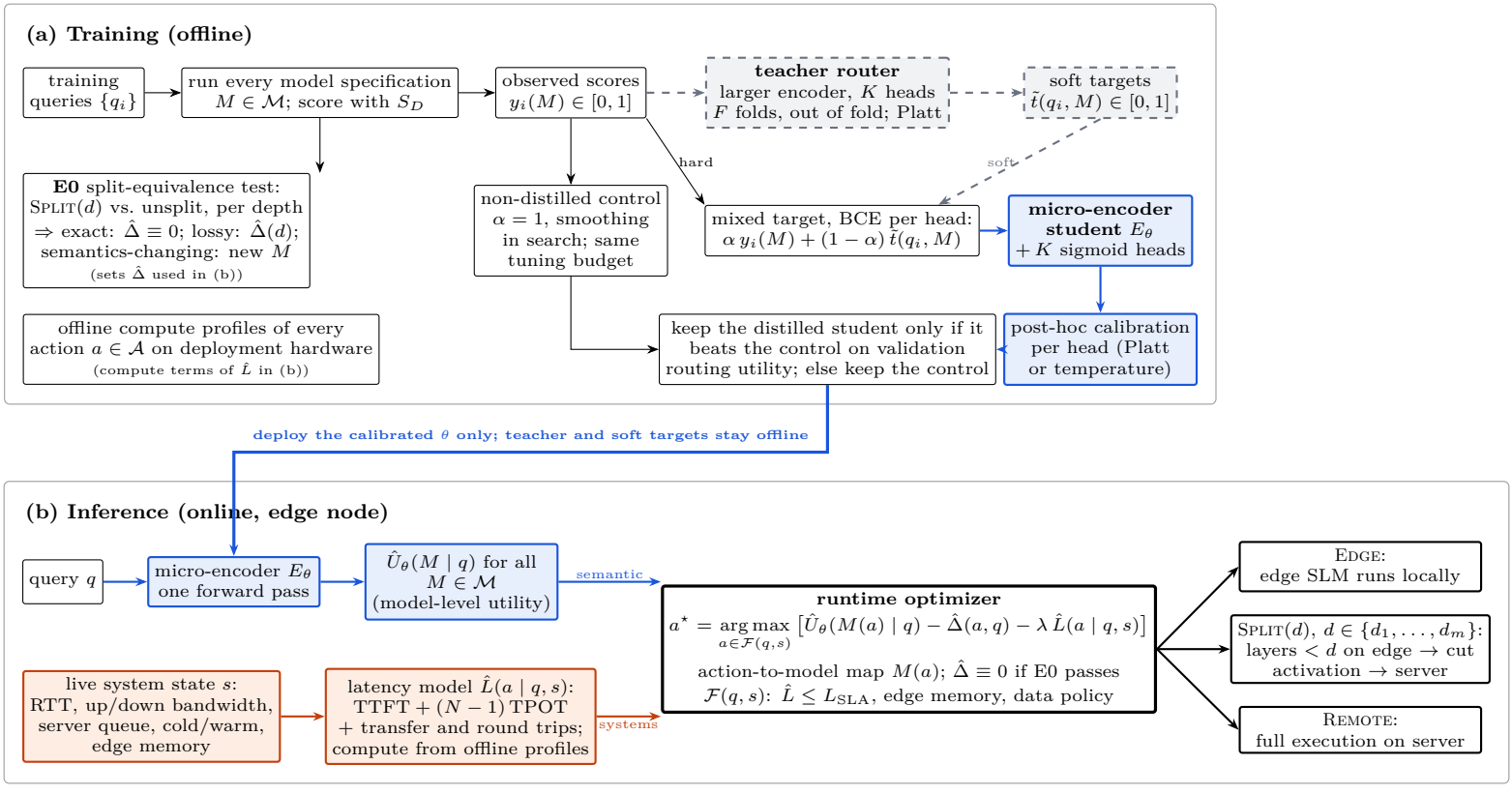
(a) Training (offline)
training queries $\{q_i\}$
run every model specification $M \in \mathcal{M}$; score with S_D
observed scores $y_i(M) \in [0,1]$
teacher router
larger encoder, K heads
F folds, out of fold; Platt
soft targets $\tilde{t}(q_i, M) \in [0,1]$
hard
soft
E0 split-equivalence test:
SPLIT(d) vs. unsplit, per depth
$\Rightarrow$ exact: $\hat{\Delta} \equiv 0$; lossy: $\hat{\Delta}(d)$;
semantics-changing: new M
(sets $\hat{\Delta}$ used in (b))
non-distilled control
$\alpha = 1$, smoothing
in search; same
tuning budget
mixed target, BCE per head:
$\alpha\, y_i(M) + (1-\alpha)\, \tilde{t}(q_i, M)$
micro-encoder student E_θ
$+ K$ sigmoid heads
offline compute profiles of every
action $a \in \mathcal{A}$ on deployment hardware
(compute terms of $\hat{L}$ in (b))
keep the distilled student only if it
beats the control on validation
routing utility; else keep the control
post-hoc calibration
per head (Platt
or temperature)
deploy the calibrated θ only; teacher and soft targets stay offline
(b) Inference (online, edge node)
query q
micro-encoder E_θ
one forward pass
$\hat{U}_\theta(M \mid q)$ for all
$M \in \mathcal{M}$
(model-level utility)
semantic
runtime optimizer
$a^\star = \arg\max_{a \in \mathcal{F}(q,s)} \left[\hat{U}_\theta(M(a) \mid q) - \hat{\Delta}(a,q) - \lambda\, \hat{L}(a \mid q, s)\right]$
action-to-model map $M(a)$; $\hat{\Delta} \equiv 0$ if E0 passes
$\mathcal{F}(q,s)$: $\hat{L} \le L_{\mathrm{SLA}}$, edge memory, data policy
live system state s:
RTT, up/down bandwidth,
server queue, cold/warm,
edge memory
latency model $\hat{L}(a \mid q, s)$:
TTFT $+ (N-1)$ TPOT
+ transfer and round trips;
compute from offline profiles
systems
EDGE:
edge SLM runs locally
SPLIT(d), $d \in \{d_1, \ldots, d_m\}$:
layers $< d$ on edge $\rightarrow$ cut
activation $\rightarrow$ server
REMOTE:
full execution on server
learned and deployed (θ)
measured or computed at run time
training only, not deployed
data, procedure or execution action
No quantity in this figure is a measured result.